
Attention Is All You Need

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NeurIPS 2017

A new state of the art — without recurrence

| **Headline result**

The Transformer reaches **28.4 BLEU** on WMT'14 English–German, a new state of the art, trained at a fraction of the cost of prior models.

- Built **only** from attention — no recurrence, no convolution.

Recurrence forces sequential computation

- RNNs compute hidden state h_t from h_{t-1} , one step at a time.
- This sequential chain blocks parallelization within a sequence.
- Long-range dependencies must traverse many steps to connect.

The limitation

Sequential computation is the fundamental constraint — and it persists even with factorization and conditional-computation tricks.

Attention is all you need

| The idea

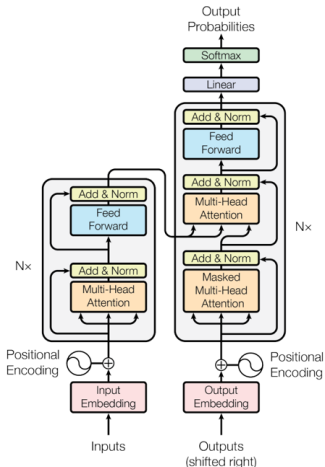
Replace recurrence and convolution **entirely** with attention, so any two positions are related in a constant number of steps.

- More computation can be parallelized during training.
- A shorter path connects every pair of positions.

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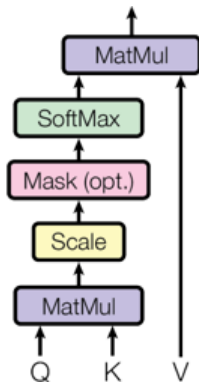
The Transformer

An encoder-decoder built from attention



- Encoder and decoder: $N = 6$ identical layers each.
- Every layer: multi-head attention + a position-wise feed-forward network.
- Residual connections + layer norm;
 $d_{\text{model}} = 512$.

Scaled dot-product attention



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

- Dot-product attention — fast and memory-efficient.
- Scale by $1/\sqrt{d_k}$ to keep softmax gradients healthy.

Many heads, many subspaces

- Project Q, K, V into $h = 8$ heads and run attention in parallel.
- Each head attends to a different representation subspace.
- $d_k = d_v = d_{\text{model}}/h = 64$, so the total cost $\approx$ one full head.

Putting order back in

- No recurrence means the model has no notion of position.
- Add **sinusoidal positional encodings** to the input embeddings.
- Different frequencies let the model attend by relative position.

Why self-attention

Layer type	Complexity / layer	Sequential ops	Max path length
Self-attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$

Why it matters

Constant path length and $O(1)$ sequential operations make long-range dependencies easier to learn, and the layer fully parallel.

How it was trained

- WMT'14 English–German (4.5M pairs) and English–French (36M).
- 8 NVIDIA P100 GPUs: base model in 12 hours, big model in 3.5 days.
- Adam with a warmup-then-decay learning-rate schedule.
- Residual dropout and label smoothing ($\epsilon_{ls} = 0.1$).

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Results

New state of the art on WMT'14

Translation quality

- English–German: **28.4 BLEU** — +2.0 over the best prior, including ensembles.
- English–French: **41.8 BLEU** — a new single-model state of the art.
- Achieved at a small fraction of the prior training cost.

Better BLEU at a fraction of the cost

Model	EN-DE	EN-FR	Train cost (FLOPs)
GNMT + RL	24.6	39.92	$1.4 \cdot 10^{20}$
ConvS2S	25.16	40.46	$1.5 \cdot 10^{20}$
Transformer (base)	27.3	38.1	$3.3 \cdot 10^{18}$
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$

Not just translation

| English constituency parsing

With little task-specific tuning, the Transformer reaches **92.7 F1** on the WSJ test set, beating most prior models.

- Strong even in the small-data, WSJ-only regime.

Attention is enough

| Takeaways

- **Simpler**: no recurrence, no convolution.
- **More parallel**: significantly faster to train.
- **Stronger**: a new state of the art in translation.

A simple template the field can build on.

Thank you

Questions welcome

